

1 Summary

Model (Run-name): model1 (test)

Iteration: 17

TAC: 154820.78839632616

Took: 2.31441330909729s

Total Time: 24.983908653259277s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

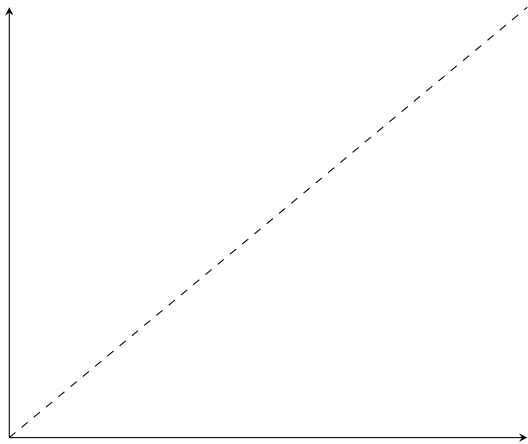
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

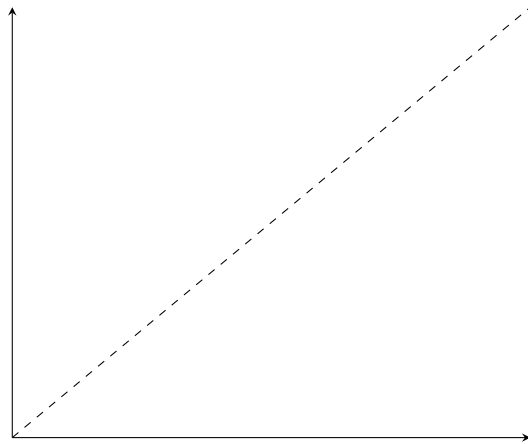
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052299746794705416	$b_{1,1,1}^{H,out}$	895.4672946654139
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01760686237913616	$b_{1,2,2}^{H,out}$	3399.124095329035
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.029307381982280722	$b_{2,1,2}^{H,out}$	4092.744255638756
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.01457543419641736	$b_{1,1,1}^{C,in}$	8358.867421861123
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009587146987906296	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022565763203393742	$b_{2,1,2}^{C,in}$	4186.326960818811
$A_{1,1,1}^\beta$	70.60541450524272	$q_{1,1,1}$	677.6509256927457	$b_{1,1,1}^{C,out}$	9036.518347553869
$A_{1,2,2}^\beta$	68.66676327863097	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	142.4338700260237	$q_{2,1,2}$	2433.5236114608815	$b_{2,1,2}^{C,out}$	6619.850572279693
$A_{cu,1}^\beta$	4.974201802143407	$q_{cu,1}$	172.34907430725434		
$A_{cu,2}^\beta$	37.6551299822246	$q_{cu,2}$	1966.4763885391185		
$A_{hu,1}^\beta$	13.213749911872785	$q_{hu,1}$	488.82546284637283		
$A_{1,1,1}$	70.60541450524272	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	68.66676327863097	$t_{1,2}^{(H)}$	582.2349074307255		
$A_{2,1,2}$	142.4338700260237	$t_{1,3}^{(H)}$	387.23490743072546		
$A_{cu,1}$	4.974201802143407	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.6551299822246	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.213749911872785	$t_{2,3}^{(H)}$	468.32381942695594		
$dt_{1,1,1}$	32.588364189758174	$t_{1,1}^{(C)}$	617.4116358102418		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	572.2349074307255		
$dt_{1,2,2}$	82.23490743072543	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	37.23490743072543	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	17.76509256927457	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.32381942695591	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	67.23490743072543	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.32381942695594	$t_{1,2,2}^H$	370.0		
$dt_{hu,1}$	62.58836418975819	$t_{2,1,2}^H$	370.0		
$f_{1,1,1}^H$	2.4201818774740915	$t_{1,1,1}^C$	618.6342496050581		
$f_{1,2,2}^H$	9.186821879267661	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	11.061470961185826	$t_{2,1,2}^C$	648.3341506855953		
$f_{1,1,1}^C$	14.60720668686363	$b_{1,1,1}^{H,in}$	1573.1182203581595		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5349.124095329034		
$f_{2,1,2}^C$	10.21055356297271	$b_{2,1,2}^{H,in}$	6526.267867099637		

3 *RecLMTD* Tangent Points

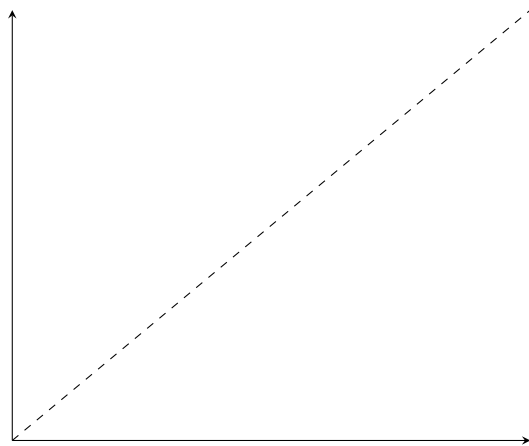
- $(1, 1, 1)$
 - Value: 0.052300
 - Error: 0.000000198214426
 - Relative error: 0.000003789954771
- $(1, 2, 2)$
 - Value: 0.017607
 - Error: 0.000000541697481
 - Relative error: 0.000030765323442
- $(2, 1, 2)$
 - Value: 0.029307
 - Error: 0.000002589712861
 - Relative error: 0.000088356034191
- $cu, 1$
 - Value: 0.014575
 - Error: 0.000000089973802
 - Relative error: 0.000006172937669
- $cu, 2$
 - Value: 0.009587
 - Error: 0.000000008186484
 - Relative error: 0.000000853901228
- $hu, 1$
 - Value: 0.022566
 - Error: 0.000000021299378
 - Relative error: 0.000000943879348



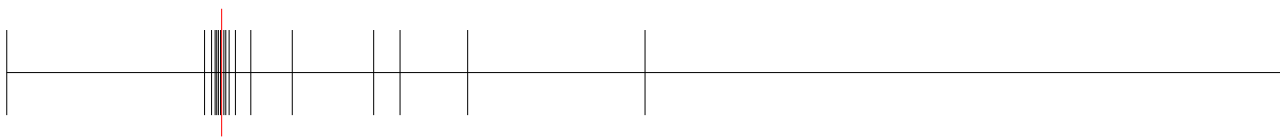
(a) *RecLMTD* - (1, 1, 1)



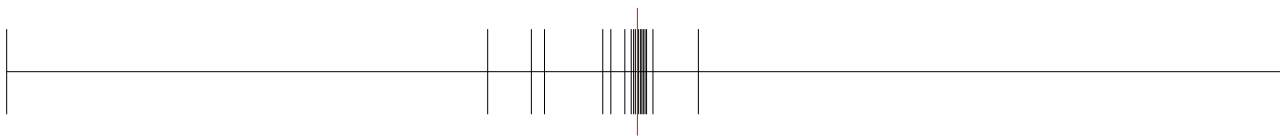
(b) *RecLMTD* - (1, 2, 2)



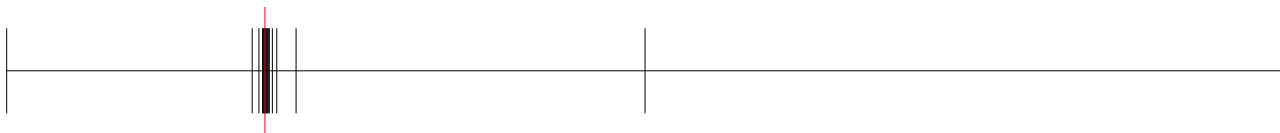
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1573.1182203581595
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5349.124095329034
 - Absolute error: 0.23570887106416194
 - Relative error: 4.4066889049492426e-05
- (2, 1, 2)
 - Value: 6526.267867099637
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 895.4672946654139
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3399.124095329035
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 4092.744255638756
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.1111061979011723e-16

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8358.867421861123
 - Absolute error: 0.1138555822399212
 - Relative error: 1.3621119624729765e-05
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4186.326960818811
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9036.518347553869
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6619.850572279693
 - Absolute error: 1.8189894035458565e-12
 - Relative error: 2.7477801555865745e-16



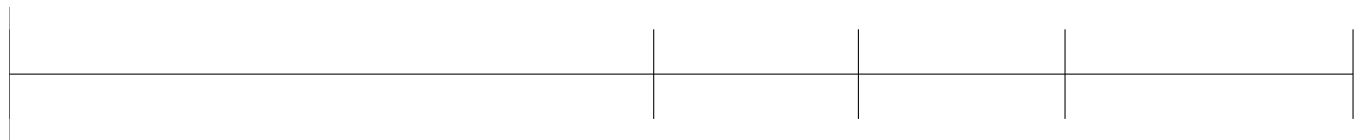
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



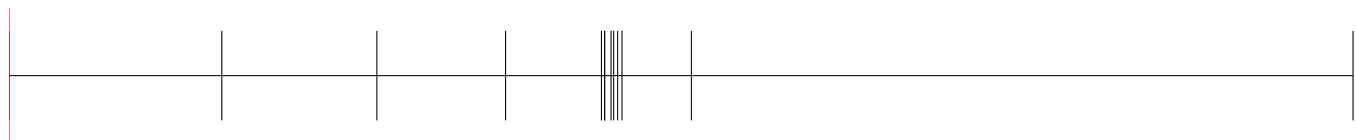
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



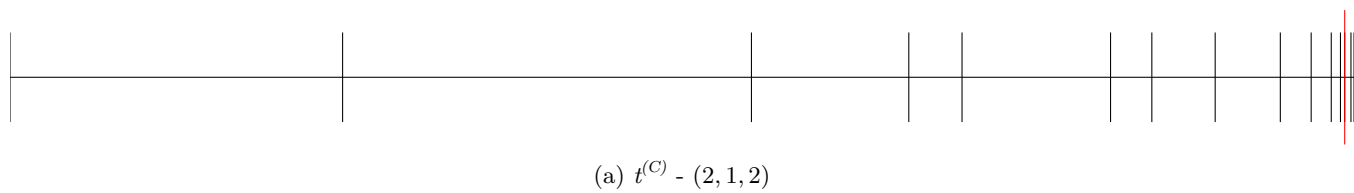
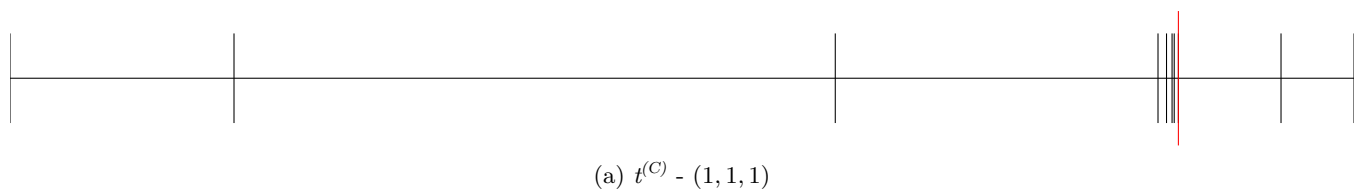
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

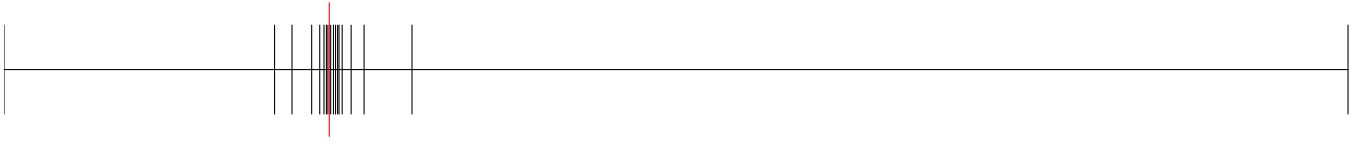
- $(1, 1, 1)$
 - Value: 70.605415
 - Error: 0.276529152613932
 - Relative error: 0.003901263683580
- $(1, 2, 2)$
 - Value: 68.666763
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 142.433870
 - Error: 0.206542061942997
 - Relative error: 0.001447991203332

5.2 A_{CU}

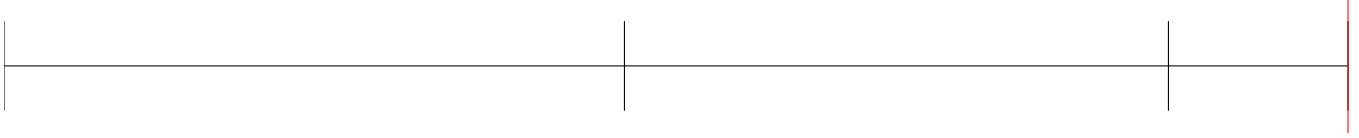
- $cu, 1$
 - Value: 4.974202
 - Error: 0.049923380614256
 - Relative error: 0.009936731032417
- $cu, 2$
 - Value: 37.655130
 - Error: 0.050666388118721
 - Relative error: 0.001343729426136

5.3 A_{HU}

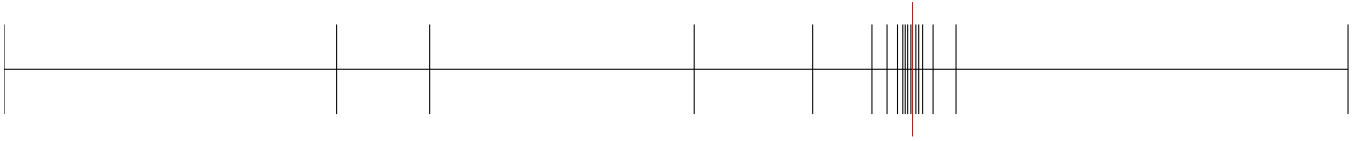
- $hu, 1$
 - Value: 13.213750
 - Error: 0.023113658983927
 - Relative error: 0.001746158284415



(a) $A^\beta - (1, 1, 1)$



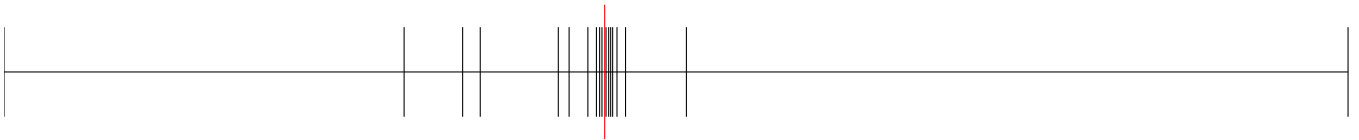
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

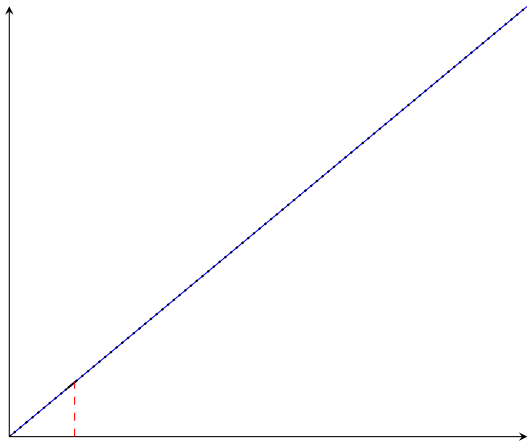


(a) $A^\beta - hu, 1$

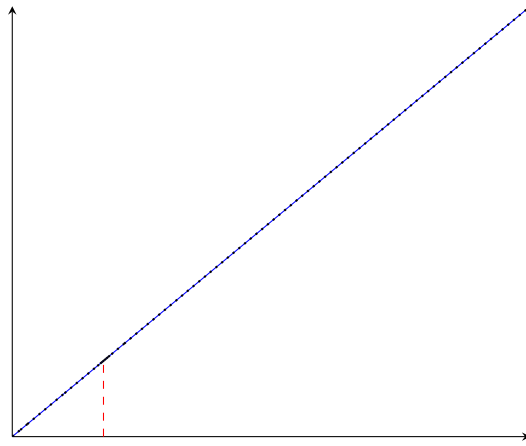
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.605415
 - $\hat{q}^\beta$: 70.605415
 - q^β : 70.605415
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, **70.50425945122109**, **71.22933662920877**, 71.2616536557244, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 68.666763
 - $\hat{q}^\beta$: 68.666763
 - q^β : 68.666763
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, **67.91344539650363**, **69.0314457108091**, 69.34502840170047, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 142.433870
 - $\hat{q}^\beta$: 142.433870
 - q^β : 142.433870
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, **140.7008934199032**, **144.70843093515643**, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.974202
 - $\hat{q}^\beta$: 4.974202
 - q^β : 4.974202
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, **4.849719301830286**, **4.983938626310074**, 4.993068227518925, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.655130

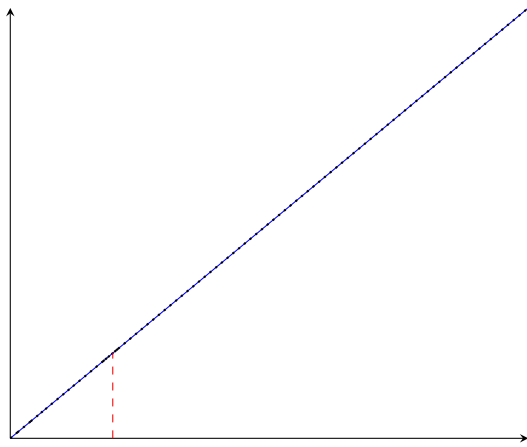
- $\hat{q}^\beta$: 37.655130
- q^β : 37.655130
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, **37.554741031910254**, **37.682669717644174**, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.213750
 - $\hat{q}^\beta$: 13.213750
 - q^β : 13.213750
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, **13.192252946941991**, **13.218222920629291**, 13.22717920580809, 13.244985069124171, 13.30871858348944, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



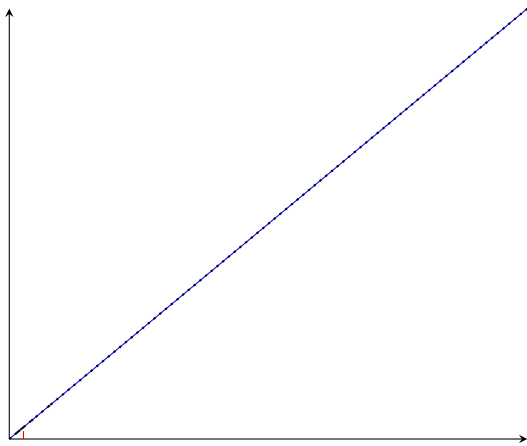
(a) $q^\beta - (1, 1, 1)$



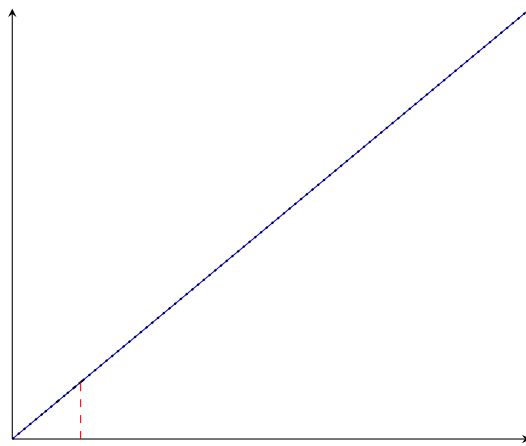
(b) $q^\beta - (1, 2, 2)$



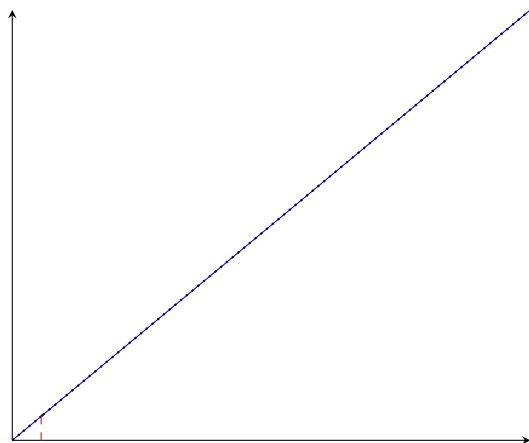
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	13.06	$b_{2,2,2}^{H,out}$	5273.732132900363
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.06	$b_{1,1,2}^{C,in}$	1963.673039181189
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.16	$b_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.016070473816068356	$b_{2,1,1}^{C,in}$	224.65618959975762
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.16	$b_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038186	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.16	$b_{1,1,2}^{C,out}$	1963.673039181189
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.0055493298268132556	$b_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.06	$b_{2,1,1}^{C,out}$	224.65618959975762
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.06	$b_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.06	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	559.5067309518668		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	590.0		
$dt_{1,2,1}$	280.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	220.0	$t_{2,1,1}^C$	500.75674797155534		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	473.2249789782197		
$f_{1,1,2}^H$	0.8131781207323385	$b_{1,2,1}^{H,in}$	4926.881779641841		
$f_{1,2,1}^H$	7.579818122525909	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	5273.732132900363		
$f_{2,2,2}^H$	8.938529038814174	$b_{1,1,2}^{H,out}$	473.2249789782197		
$f_{1,1,2}^C$	4.789446437027291	$b_{1,2,1}^{H,out}$	4926.881779641841		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.39279331313636945	$b_{2,2,1}^{H,out}$	0.0		